

Matthew Coming

Austin, Texas

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EDUCATION

Texas A&M, College of Engineering

Bachelor of Science in Computer Science; GPA: 3.7

Minor in CyberSecurity

College Station, Texas

Aug 2017 – May 2021

SKILLS

- **Languages:** C, C++, C#, Python, Javascript, Bash
- **Software Specialties:** Docker, Gitlab CI/CD, OpenGL, React
- **Technologies:** Linux, Kafka, AWS S3, NASA GMAT

EMPLOYMENT HISTORY

• York Space Systems

Austin, Texas

Senior Software Engineer II

August 2023 – Present

- My role has been to design, build, and support software across the entire software organization to support the Proliferated Warfighter Space Architecture Traunch 1 and Traunch 2 launches, in addition to other missions.
- **Ground Software (GSW)**
 - * Designed York's original GSW-integrated flight dynamics solver.
 - * Led integration and test of York's GSW against the US Government's GSW.
 - * Designed and led development of multiple internal testing tools to aid QA test development.
 - * Created CI/CD pipelines across products to allow for continous testing, building, and semantic versioning internal dependency management.
- **Simulation, Integration, and Test**
 - * Designed York's GSW integration test environment, allowing for repeatable tests of the GSW against York's end-to-end vehicle simulation.
 - * Productized and proliferated York's end-to-end vehicle simulation across the organization.
 - * Packaged an offline runtime environment of the end-to-end vehicle simulation for work in secure spaces.
 - * Manage a custom linux kernel for windows subsystem for linux (WSL) development, allowing for development not possible in WSL out of the box.
 - * Designed and built a vehicle camera simulation for flight-like testing of video processing using OpenGL.

• Emergent Space Technologies

Austin, Texas

Software Engineer

June 2021 – August 2023

- Built stateless microservices, deployed in a RedHat OpenShift environment, to aid Space Domain Awareness operations.
- **Microservice Design:** Designed and implemented docker images, message serialization, deserialization, and validation to communicate across a pub/sub message bus.
- **GMAT:** Automated orbit determination, propagation, and satellite visibility calculations using NASA GMAT.
- **Python:** Made extensive use of python 3.7+ packaging, Metaclasses, custom annotations, multiple inheritance, and polymorphism.

• L3Harris Technologies

Mission Integration Division, Greenville, Texas

Co-op Software Engineer

3 terms, 11 Months

- Member of a team responsible for maintaining, refactoring, developing, and documenting software for the internal research and development of intelligence and reconnaissance systems, primarily mission management software.
- **JavaScript and React:** Designed and wrote ReactJS UI components that communicated with a backend message bus through Google Protobuf messages.
- **C# and WinForms:** Designed and wrote application code to expand the capabilities of L3Harris ViewPoint HD.
- **Security:** Used Fortify static code analysis to triage and fix security defects in a legacy code base.
- **GIS:** Worked extensively with CesiumJS, an open source JavaScript library for creating 3D globes and maps, and ESRI ArcMap.

PROJECTS AND INTERESTS

- **Hobby Mechanic:** Working on restoring and upgrading a 2001 Porsche 911 inside and out.
- **OpenGL, ThreeJS:** Pursuing the creation of algorithmic art and animation in shader languages.
- **AWS:** Hosting MatthewComing.com using S3, Cloudfront, and Route 53.
- **Relevant Coursework:** Software Engineering, Software Security, Computer Graphics, Operating Systems, Machine Learning, Artificial Intelligence

ORGANIZATIONS AND CLUBS

- *TAMU AutoDrive Challenge Team - Member* *Spring 2021*
 - Real-time HD Map Verification – Using the Robot Operating System(ROS), React, CesiumJS, and Electron, this team created a system that allows a user to verify the correctness of a high definition map during the maneuvering of a self-driving car.
- *TAMU CyberSecurity Club - Member* *Fall 2018, Spring 2019*
- *TAMU BUILD - Student Supervisor* *Fall 2018, Spring 2019*
 - BUILD is a student run organization at Texas A&M University that transforms shipping containers into medical clinics and classrooms that are sent all around the world. The role of a Student Supervisor is to guide the work and oversee the safety of the hundreds of volunteers each semester.

HONORS AND CERTIFICATION

- *2x York Space Systems Company Recognition Award*
- *2x Emergent Space Technologies Company Recognition Award*
- *Texas A&M University Distinguished Student Award* *Spring 2018, Spring 2019*
- *Texas A&M University Dean's Honor Award* *Fall 2017, Fall 2018, Spring 2020*